

You-Ting Wu

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Education

Columbia University

Ph.D., Applied Mathematics

New York, NY, USA

Sept 2024 – Aug 2028 (expected)

National Taiwan University

B.Sc., Mechanical Engineering and B.Sc., Atmospheric Sciences

Taipei, Taiwan

Sept 2018 – July 2023

Skills

Machine learning: deep learning, CNN, LSTM, neural network surrogate models, out-of-sample validation

Statistics: statistical modeling and inference, regression, hypothesis testing, bootstrap resampling, variance decomposition, time series

Mathematics: nonlinear dynamical systems, numerical analysis, linear algebra, numerical PDEs

Programming: Python (PyTorch, SciPy, Scikit-learn, data visualization), C/C++, Fortran

High-performance computing: MPI, HPC clusters

Research Experience

Graduate Research Assistant, Columbia University, New York, NY, USA

Sept 2024 – Present

- Formulated and tested hypotheses on nonlinear, bifurcating dynamics and regime transitions in a complex physical system under 6 forcing scenarios, identifying a critical tipping threshold through large-ensemble numerical simulation
- Built and validated deep learning surrogate models (CNN and LSTM) in Python (PyTorch) that predict high-dimensional spatiotemporal time series over 70-year horizons with roughly 100x speedup over direct simulation; evaluated out-of-sample generalization across unseen scenarios to guard against overfitting
- Applied statistical modeling and inference (regression, t-tests, bootstrap resampling) to a 20-member ensemble spanning 2,000+ simulated years, quantifying significance and robustness of results against internal variability (noise)
- Ran and modified a large Fortran codebase parallelized with MPI on university HPC clusters to generate and process the simulation dataset for downstream analysis

Software Engineering Intern, Formosa Climate Smart Service, Taipei, Taiwan

July 2023 – Sept 2023

- Designed an ETL pipeline that transforms raw engineering inputs into quantitative Energy Use Intensity (EUI) metrics, the standard measure of energy consumption normalized by floor area, across large building datasets
- Developed a Python-based analytics and data visualization platform to compute and visualize EUI metrics across large building datasets
- Automated the integration of geospatial and CAD geometry data into a 3D building-EUI workflow, linking building geometry to quantitative energy metrics

Visiting Student, Lamont-Doherty Earth Observatory, Palisades, NY, USA

June 2022 – Aug 2022

- Applied bootstrap resampling and statistical significance testing to large-ensemble simulations to quantify the robustness of a physical signal against internal variability (noise)

Publications

- Previdi, M., **Wu, Y.-T.**, Polvani, L.M., England, M.R., Sigmond, M., Liang, Y.-C. Substantial Twentieth-Century Global Sea Level Rise Caused by Ozone-Depleting Substances. Submitted to Journal of Climate (under revision).
- Wu, Y.-T.**, Previdi, M., Romanou, A., Lerner, P., Polvani, L.M. Arctic Climate Tipping Points under Delayed CO2 Emissions Mitigation. Accepted by Communications Earth and Environment (to appear).
- Liang, Y.-C., Polvani, L.M., Previdi, M., Dong, Y., **Wu, Y.-T.**, England, M.R., Griffith, S.M. (2026). The Importance of Non-CO2 Greenhouse Gases to Arctic Warming and Sea-Ice Loss. Geophysical Research Letters, 53, e2026GL122867.
- Wu, Y.-T.**, Liang, Y.-C., Previdi, M., Polvani, L.M., England, M.R., Sigmond, M., Lo, M.-H. (2024). Stronger Arctic Amplification from Anthropogenic Aerosols than from Greenhouse Gases. npj Climate and Atmospheric Science, 7(1), 142.
- Wu, Y.-T.**, Liang, Y.-C., Kuo, Y.-N., Lehner, F., Polvani, L.M., Previdi, M., Lo, M.-H., Lan, C.-W. (2023). Exploiting SMILES and the CMIP5 Archive to Understand Arctic Climate Change Seasonality and Uncertainty. Geophysical Research Letters, 50(2), e2022GL100745.

Awards

- Government Scholarship to Study Abroad, Ministry of Education, Taiwan (June 2026)
- Nominated for membership, Sigma Xi, The Scientific Research Honor Society, USA (January 2026)
- Excellent Undergraduate Research Project Award, Ministry of Science and Technology, Taiwan (June 2023)
- Undergraduate Research Project Grant, Ministry of Science and Technology, Taiwan (June 2022)

Professional Service, Presentations, and Teaching

- Journal reviewer: Nature Communications (2026)
- 12 scientific presentations at international conferences (2021–2025)
- 12 teaching assistantships, including Partial Differential Equations, Numerical Analysis, and Applied Mathematics